MIL-E-1/147B
15 October 1970
SUPERSEDING
MIL-E-1/147A
22 April 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5A6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: **Pentode, RF power amplifier and oscillator**

Outline	---	6-3 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	a	nc	g3	-f (+par)	+f	g2	g1	g3	fct

ABSOLUTE-MAXIMUM RATINGS: (B)

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ib	(B) Pp	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	Vdc	mAdc	W	ft
Maximum:	2.9 or 5.8	150	-75	150	0	40	5	(See note 1)
Minimum:	2.1 or 4.3	---	---	---	---	---	---	---
TEST CONDITIONS:	5	150	-10	150	0	---	---	---

GENERAL:

Qualification - Not required

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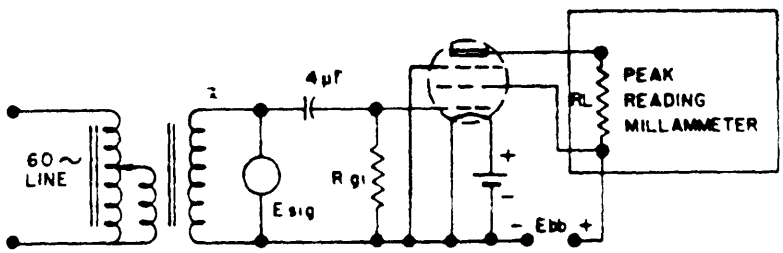
METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	See note 2	0.65	II	Ic1	---	-2.0	µAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	21	35	mAdc
1256	Electrode current (2) (anode)	Ec1 = -25 Vdc	0.65	II	Ib	---	2.0	mAdc
1256	Electrode current (screen)		0.65	II	Ic2	0.0	4.0	mAdc
----	Operation peak output current (1)	Ebb = Ec2 = 120 Vdc; Rl = 320 ohms; Ef = 5.0 Vdc; Esig = 20 Vac; Rg1 = 22,000 ohms; Ecc1 = 0 (see note 3)	0.65	II	ib	110	---	ma
----	Operation peak output current (2)	Ebb = Ec2 = 120 Vdc; Rl = 320 ohms; Ef = 4.5 Vdc; Esig = 20 Vac; Rg1 = 22,000 ohms; Ecc1 = 0 (see note 3)	0.65	II	Δib	---	10	ma
(B) 1201	Short and discontinuity detection	See note 4	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1301	Filament current		---	---	If	210	250	mA
1236	Power oscillation	F = 70 MHz; Rg1 = 15,000 ohms; Rg2 = 1,500 ohms (min) by passed by 1,000 pF (min); Esig/Ic1 = 1 mAdc; Ib = 40 mAdc	---	---	Po	2.4	---	W
1306	Transconductance		---	---	Sm	3,200	5,400	µmhos
1331	Direct-interelectrode capacitance	Without shield	---	---	Cgp	---	0.15	pF
					Cin	6.5	10.5	pF
					Cout	4.5	7.1	pF
(B) 1121	Base strain		---	---	---	---	---	---
(B) 2126	Glass strain		4.0	I	---	---	---	---
(B) 1211	Insulation of electrodes		---	---	---	---	---	---
(B) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
----	Life-test provisions	Group A; Ec1/Ib = 33 mAdc; Ef = 5.0 V	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued							
----	Life-test end points (250 hours)	Operation peak output current	---	---	ib	100	---	ma
		Power oscillation (see note 5)	---	---	Po	2.0	---	W

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Use circuit as below:



All power supplied shall have negligible impedance to operating frequency.
Grid signal impedance shall be less than 5.0 ohms; voltage sinusoidal.

4. The short and discontinuity detection test shall be conducted using the mechanical tapper, specified in method 1201, with the minimum average amplitude of the tap equivalent to 40 G peak acceleration.

(B) 5. During life, Ic1 for power oscillation test shall be 0.6 to 1.2 mAdc. Initially adjust Ec1 for Ib = 33 mAdc on a bogey tube having an anode current of 28 ± 1.4 mAdc.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL,
Navy -
Air Force - 11, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2425-19)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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